Mathematics for Computer Science – 1

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What is Mathematics for Computer Science – 1?

According to LLM17-MCS:

- Proof techniques – 证明方法
- Number Theory – 初等数论
- Abstract Algebra – 抽象代数
- Probability – 概率论
- Algorithms Analysis – 算法分析

What is Mathematics for Computer Science – 2?

According to 《Concrete Mathematics》(GKP93–《具体数学》):

- Number Theory – 初等数论
- Probability – 概率论
- Algorithms Analysis – 算法分析相关的数学

What is Mathematics for Computer Science – 2?

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本人浅见!

《计算机科学中的数学》就是《离散数学》。

Motivations of our course.

从 2024 年起，《计算机科学中的数学》第一次出现在 SCNU-CS，以下专题为本课程的主线:

- Number Theory – 初等数论
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Notice!

强调，我们关注编程与算法，即所谓的"computational(计算性)"，且以上专题将融合在一起。

参考书.

- A Concrete Introduction to Number Theory and Algebra(CINTA，本课程教材)
- Number Theory: A Friendly Introduction to Number Theory (FINT)
- Abstract Algebra: Abstract Algebra: Theory and Applications (AATA)
- Algorithms: Introduction to Algorithms (CLRS)

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Wanted!

Reading, thinking and programming.

Remark.

本课程期望的目标.

同学们可以建立起最基本、正常的数学学习思维，并多做练习.

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本课程一种重要的思路.

Finding patterns. （寻找模式.）

AI 时代的数学课特征与预警.

AI 时代的数学课有什么特征.

- 更丰富的知识；更丰富的阅读资料；
- 更唾手可得的答案；

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AI 时代的数学课需要警惕什么？

- 知识、资料过多，学习路径的选择更复杂；
- 容易得到答案，让本来就缺乏思考的大学生更难以抗拒惰性，“不思考”的状况进一步恶化；

Numbers.

Number systems

$$\mathbb{N} = \{0, 1, 2, \cdots\}$$

$$\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \cdots\}$$

$$\mathbb{Q}$$

$$\mathbb{R}$$

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$$\mathbb{Q}$$

$$\mathbb{R}$$

问题.

计算机系统中的数学结构主要是什么？

Warm-up-1.

Simple Code:

```
1 unsigned char a = 255, b = 2;
2 a = a + b;
3 printf("a == %d \n", a);
```

问题.

这段程序输出什么？

Warm-up-2.

Simple Code:

```
1  int n = 32; /*try different values here;*/
2  unsigned int val = 1; /*or try a different type here;*/
3  for(int i = 1; i <= n; i++)
4  {
5          val = val * 2;
6          printf("What you get is %u \n.", val);
7  }
```

问题.

这段程序输出什么？

Warm-up-3.

CSers 的计算方法.

请计算以下等式：

$$1 + 2 + 2^2 + 2^3 + \cdots + 2^{10}$$

问题.

这道题的意义在哪里？

Numbers in Computer Systems.

Observations:

- All computations in computers use fixed-size representations(定长计算).
- Addition（加法）is natural while multiplication（乘法）is not.
- Subtraction（减法）is the same as addition; the relation between multiplication and division（除法）is more complicated.

Numbers in Computer Systems.

Observations:

- All computations in computers use fixed-size representations(定长计算).
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注意!

以下我们重点关注计算机系统中的加、减、乘、除.

Simple rules for addition and multiplication.

Well-known and useful rules of arithmetic:

- The last bit of an even(odd) number is $0(1)$;
- To multiply(divide) an unsigned integer by 2 is equivalent to shifting left(right) the number by 1 bit.

诡异的加法.

整数加法

```
1  //输入: 两个整数a和b
2  //输出: a与b的和
3  int add(int a, int b) {
4      if (b == 0) return a;
5      return add(a ^ b, (b & a) << 1);
6  }
```

问题.

如何理解这个算法？

理解加法.

要理解这个加法算法只需要正确回答出以下三个问题：

1. $a \wedge b$ 得到的是什么？
2. $(b \& a) << 1$ 得到的是什么？
3. 该算法为什么会终止？

乘法的学习目标.

学习目标.

本节的主要内容：朴素乘法、简单乘法（递归版本与迭代版本）。建立算法分析的基本思路。

Naïve Multiplication.

阅读并思考以下代码，理解它的计算方法。

```
1  # Input: two integers a and b, where a >= b.
2  # Output: the product of a and b.
3  def naive_multiply(a, b):
4      if b == 0:
5          return 0
6      return a + naive_multiply(a, b - 1)
```

问题.

以上算法是否正确？效率如何？

Three key issues.

In general, three key issues should be considered when we encounter a new algorithm.

- **Correctness(正确性)**. Does the algorithm correctly achieve the goal?
- **Efficiency(效率、复杂度)**. How long does the algorithm take when it terminates?
- **Optimization(优化)**. Can we provide a better method?

Rule of Multiplication.

为了得到计算 $a * b$ 的优化算法，先思考以下递归的计算过程。

Rule of Multiplication.

$$a \cdot b = \begin{cases} 2(a \cdot \lfloor b/2 \rfloor) & \text{if } b \text{ is even;} \\ a + 2(a \cdot \lfloor b/2 \rfloor) & \text{if } b \text{ is odd.} \end{cases} \tag{1}$$

Simple multiplication 的伪代码.

```
1  # Input: two integers a and b.
2  # Output: the product of a and b.
3  def multiply(a, b):
4      if b == 0:
5          return 0
6      if is_even(b): # the last bit of b is 0;
7          return 2*multiply(a, b//2)
8      else: # b is odd
9                  return 2*multiply(a, b//2) + a
```

问题.

以上算法是否正确？效率是否提高？

Simple multiplication 的迭代版本.

如何得到一种迭代式的乘法算法？考虑以下表达式：

Intuition.

For two n-bits integers a and b, view b as a binary number in position notation(位置记数法).

$$\begin{aligned} a \cdot b &= a \cdot (b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \cdots + b_1 2 + b_0) \\ &= b_{n-1}(a \cdot 2^{n-1}) + b_{n-2}(a \cdot 2^{n-2}) + \cdots + b_1(a \cdot 2) + b_0 a \end{aligned}$$

Observations: 1. Every term in the last equation has $2^i a$, which means we must continue to do $a* = 2$
2. The bit b_i in every term plays as a flag, when $b_i = 1$, we must have $res+ = a$;

Simple multiplication 的迭代版本.

位置计数法给我们的启发：

观察.

- Every term in the last equation has $2^i a$, which means we must continue to do $a* = 2$
- The bit b_i in every term plays as a flag, when $b_i = 1$, we must have $res+ = a$;

Simple multiplication-计算实例.

Example.

To compute 3×10.

Simple multiplication-计算实例.

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10's binary string is 1010.

Simple multiplication-计算实例.

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Repeatedly multiply 3 by 2^i and get: 3, 6, 12, 24.

Simple multiplication-计算实例.

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To compute 3×10.
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Choose the right terms to add, they are 6 and 24.

Simple multiplication-计算实例.

Example.

To compute 3×10.
10's binary string is 1010.
Repeatedly multiply 3 by 2^i and get: 3, 6, 12, 24.
Choose the right terms to add, they are 6 and 24.
Return the result which is 30.

Simple multiplication.

Analysis

- Correctness. Obviously.
- Efficiency. $\mathcal{O}(n^2)$ bit operations. Why?
- Optimization. To be or not to be optimized? Why?

Notice!

The multiplication is implemented *without* using multiplication !

Remarks.

思考.

请不要把这里的“加、减、乘、除”仅仅视为计算机系统中需要实现的计算方法，而应该把它们理解为某种“模式”、某种“思维方式”。请在后续课程中不断体会。

练习题.

对比以下两种数列求积算法，说明并验证它们的效率。

```
def product1(X):
    res = 1
    for i in range(len(X)):
        res *= X[i]
    return res
```

```
def product2(X):
    if len(X) == 0: return 1
    if len(X) == 1: return X[0]
    if len(X) == 2: return X[0] * X[1]
    l = len(X)
    return product2(X[0:(l + 1) // 2]) * product2(X[(l + 1) // 2: l])
```

除法.

本节学习目标.

本节的主要内容：除法算法、良序公理、除法算法的证明。

Division Algorithm.

Theorem

(Division Algorithm) Given integers a and b, with $b > 0$, there exist unique integers q and r, called quotient（商）and remainder（余数）respectively, such that

$$a = qb + r$$

where $0 \leq r < b$.

Remark.

除法算法并不仅仅是算法或定理。

Geometric interpretation of division.

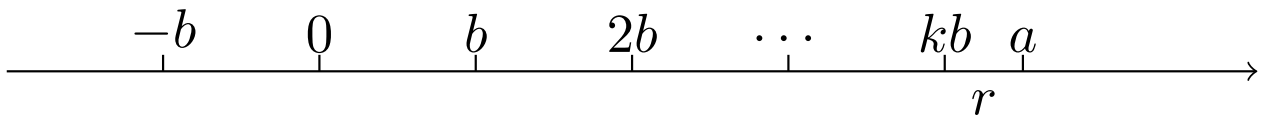

Figure: Geometric interpretation of division.

Remark.

除法算法的几何解释能在后续帮助我们设计算法，请留意。

Axiom.

在证明除法算法之前，先解释一种常用公理。

Principle of Well-Ordering(良序原则).

Every nonempty subset of natural numbers contain a least element.

Proof of division theorem.

The proof contains two parts: existence(存在性) and uniqueness(唯一性).

Proof.

- **Existence of q and r.** 证明存在性使用什么方法？

Proof of division theorem.

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Proof.

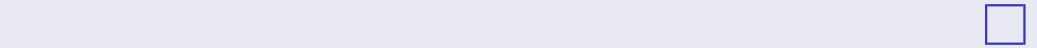
- **Existence of q and r.** 证明存在性使用什么方法？
  构造法：Given a and b, construct a set

$$S = \{a - bk : k \in \mathbb{Z} \text{ and } (a - bk) \geq 0\}.$$

Proof of division theorem（Continue）.

Proof.

- It is easy to show that S is nonempty. By the **Principle of Well-Ordering**, there is a smallest member $r \in S$, and $r = a - qb$. Therefore, $a = qb + r$, $r \geq 0$. To show $r < b$ is left as an exercise.
- **Uniqueness of q and r.** This is an exercise. [提示：证明唯一性使用什么方法？反证法！]



Naïve Division.

算法的实现：

```
1  def naive_divide(a, b):
2      q, r = 0, 0
3      while(a >= b):
4          a, q = a - b, q + 1
5      r = a
6      return q, r
```

问题.

算法是否正确？效率如何？能否优化？

Simple Division.

```
1  def divide(a, b):
2      if(a == 0):
3          return 0, 0
4      q, r = divide(a // 2, b)
5      q, r = 2 * q, 2 * r
6      if(a & 1): #if a is an odd number
7          r = r + 1
8      if(r >= b):
9          r, q = r - b, q + 1
10     return q, r
```

Simple Division-思考.

问题.

算法是否正确？效率如何？是否对朴素除法的优化？为什么？

Important remark.

重要提示.

Multiplication can be implemented using only addition and shifts, while division can be implemented using only subtraction and shifts. More importantly, the point is not merely implementation, but a way of thinking that will reappear throughout the course.

欧几里得算法.

以下我们转入一个新的主题：欧几里得算法。先介绍一些定义与概念。

- **Divisibility.** Let a and b be integers, a is said to be *divisible* by b when $a = qb$ for some integer q, we write $b \mid a$.
- **Common divisor.** An integer d is called a common divisor of a and b if $d \mid a$ and $d \mid b$.
- **Greatest common divisor.** An integer d is the greatest common divisor($\gcd$) of a and b, if d is a common divisor of a and b, and for any other common divisor of a and b, say d', then $d' \mid d$.
- **Relatively prime.** If $\gcd(a, b) = 1$, we say that a and b are relatively prime.

The Euclidean Algorithm.

Euclidean algorithm(gcd algorithm).

Given two positive integers a and b with $a \geq b$:

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

where $a \bmod b$ means to divide a by b to get the remainder r.

The Euclidean Algorithm.

一个计算实例。

Example

Given $a = 12345$ and $b = 678$,

$$\begin{aligned}\gcd(12345, 678) &= \gcd(678, 141)\\ &= \gcd(141, 114)\\ &= \gcd(114, 27)\\ &= \gcd(27, 6)\\ &= \gcd(6, 3)\\ &= \gcd(3, 0)\end{aligned}$$

Hence, $\gcd(12345, 678) = 3$.

The Euclidean Algorithm.

gcd 算法的递归实现如下：

```
1  # Input: positive integers a and b, with a > b .
2  # Output: the greatest common divisor of a and b.
3  def gcd(a, b):
4      if(b == 0):
5          return a
6      else:
7          return gcd(b, a % b)
```

Correctness of The Euclidean Algorithm.

Correctness. It is enough to show a slightly simpler rule:

$$\gcd(a, b) = \gcd(a - b, b).$$

From this rule, we can repeatedly subtract b from a (recall the Naïve division algorithm) and derive the final rule as follows:

$$\begin{aligned}\gcd(a, b) &= \gcd(a - b, b) \\ &= \gcd((a - b) - b, b) \\ &= \cdots \\ &= \gcd(a \mod b, b).\end{aligned}$$

Correctness of The Euclidean Algorithm.

To show $\gcd(a, b) = \gcd(a - b, b)$, we have two directions to prove.

- any integer that divides both a and b must also divides $a - b$, thus $\gcd(a, b) \leq \gcd(a - b, b)$.
- any integer that divdes both $a - b$ and b must also divide a and b, thus $\gcd(a - b, b) \leq \gcd(a, b)$.

Efficiency of The Euclidean Algorithm.

Efficiency. To understand that the gcd algorithm is quite fast, we should be aware of the fact that $a \mod b$ is a substantial reduction. It is easy to check:

Lemma

If $a \geq b$, then $(a \bmod b) < a/2$.

Hence, after two consecutive rounds, both a and b are at least halved in value, which means that the algorithm will terminate after $\mathcal{O}(\log a)$ recursive calls.

Optimization of The Euclidean Algorithm.

Can we do better? The *binary gcd algorithm*, also known as Stein's algorithm, has the same function as the gcd algorithm. This algorithm avoids complex operations, such as division and multiplication; instead, it relies only on subtraction, and more efficient bit right shift and bit left shift operations. The analysis of binary gcd algorithm is an exercise.

Bézout's Theorem.

Theorem

(Bézout) Let a and b be nonzero integers. Then there exist integers r and s such that

$$\gcd(a, b) = ar + bs$$

Furthermore, the greatest common divisor of a and b is unique.

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Furthermore, the greatest common divisor of a and b is unique.

Proof of Bézout's Theorem– 提示 1.

Proof.

构造集合

$$S = \{am + bn : m, n \in \mathbb{Z} \text{ 且 } am + bn > 0\}.$$

显然，集合 S 非空，根据良序原则，取其中最小值 $d = ar + bs$。然后证明两个属性：d 是 a 和 b 的公因子（提示：除法算法！）；如果存在 a 和 b 的公因子 d'，则 $d' \mid d$。这样就证明了 $d = \gcd(a, b)$。 □

Proof of Bézout's Theorem– 提示 2.

Proof.

证明 d 整除 a。根据除法算法，记 $a = dq + r'$，$0 \leq r' < d$。证明 $r' = 0$。如果 $r' > 0$，则有：

$$\begin{aligned} r' &= a - dq \\ &= a - (ar + bs)q \\ &= a(1 - qr) + b(-qs) \end{aligned}$$

所以，r' 也落在集合 S 中，但是，这与 d 是集合 S 中最小元素矛盾。因此，r' 只能是 0。同理，可证 d 整除 b。请读者完成余下细节，留作练习。 □

Bézout's Theorem 的应用.

Bézout 定理的证明还显示这样一个重要的事实：

> $\gcd(a, b)$ 等于 a 和 b 所有线性组合值的最小值。

记住这个事实，它可经常帮助我们解题。比如，我们立即可得以下推论。

Corollary

整数 a 和 b 互素当且仅当存在整数 r 和 s 使得 $ar + bs = 1$。

Bézout's Theorem 的应用.

Theorem

设 $a, b \in \mathbb{Z}$，p 为素数。如果 $p \mid ab$，则 $p \mid a$ 或 $p \mid b$。

Bézout's Theorem 的应用.

Theorem

设 $a, b \in \mathbb{Z}$，p 为素数。如果 $p \mid ab$，则 $p \mid a$ 或 $p \mid b$。

Proof.

假设 $p \nmid a$，证明 $p \mid b$。因为 $p \nmid a$ 且 p 是素数，则 $\gcd(a, p) = 1$，即存在 $r, s \in \mathbb{Z}$ 使得 $ar + ps = 1$。因此，

$$b = b(ar + ps) = abr + pbs.$$

既然 $p \mid ab$，那么 p 同时整除 abr 和 pbs，则 p 必然整除 b. □

Bézout's Theorem 的应用.

Corollary

设 $a, b, n \in \mathbb{Z}$，$n > 0$。如果 $n \mid ab$ 且 $\gcd(a, n) = 1$，则 $n \mid b$.

Proof.

使用以上类似的方法，易得，留作练习。 □

The Extended Euclidean Algorithm.

The extended Euclidean algorithm.

The *extended Euclidean algorithm*(egcd-algorithm) is a simple extension of the Euclidean algorithm. Given two nonzero integers a and b, it efficiently computes the *Bézout coefficients* r and s, and plays an important role in modular arithmetic.

Example

Given $a = 19$, $b = 13$, to find r and s, such that $ar + bs = \gcd(19, 13)$.

$$a \cdot 1 + b \cdot 0 = 19 \tag{2}$$

$$a \cdot 0 + b \cdot 1 = 13 \tag{3}$$

$$a \cdot 1 + b \cdot (-1) = 6 \tag{4}$$

$$a \cdot (-2) + b \cdot 3 = 1 \tag{5}$$

Finally, we have $r = -2$, $s = 3$ and $d = 1$, which is the answer.

The Extended Euclidean Algorithm.

Recapitulate the process.

- Write out two obviously true linear equations.
- Repeatedly subtract some integer multiple of one equation from the other, until the right-hand side of one equation achieves $\gcd(a, b)$.

The Extended Euclidean Algorithm.

Two Observations.

At least two facts should be noticed

- The right-hand sides of these equations undergo the same iterations as in the Euclidean algorithm.
- The a and b are just place holders, they are not involved in the computation, we can safely get rid of them in the equations.

例子

Given $a = 13$, $b = 9$, to find r and s, such that $ar + bs = \gcd(13, 9)$. We write out a matrix:

$$\begin{pmatrix} 1 & 0 & 13 \\ 0 & 1 & 9 \end{pmatrix}$$

If we subtract row 2 from row 1, we get a new matrix:

$$\begin{pmatrix} 0 & 1 & 9 \\ 1 & -1 & 4 \end{pmatrix}$$

例子-续

Then we subtract 2 multiple of row 2 from row 1, we get:

$$\begin{pmatrix} 1 & -1 & 4 \\ -2 & 3 & 1 \end{pmatrix}$$

Finally, we have $r = -2$, $s = 3$ and $d = 1$, which is the answer.

The Extended Euclidean Algorithm.

From the perspective of matrix, the process starts with a matrix

$$M = \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \end{pmatrix}$$

which satisfies

$$M \begin{pmatrix} a \\ b \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

By repeatedly subtracting a multiple of one row of M from the other row or exchanging the two rows, finally we obtain a matrix M'

$$M' = \begin{pmatrix} r & s & d \\ R & S & 0 \end{pmatrix}$$

The Extended Euclidean Algorithm.

$$M' = \begin{pmatrix} r & s & d \\ R & S & 0 \end{pmatrix}$$

satisfies

$$M' \begin{pmatrix} a \\ b \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

That means $ar + bs = d$.

课堂练习.

Given $a = 252$, $b = 198$, to find r, s and $d = \gcd(a, b)$, such that $ar + bs = d$.

The Extended Euclidean Algorithm.

The extended Euclidean algorithm is essentially the process of computing Bézout coefficients we described above.

```
# Input: integers a and b, with a > b .
# Output: d=gcd(a,b), and
# r and s s.t. d = a*r + b*s
def egcd(a, b):
    r0, r1, s0, s1 = 1, 0, 0, 1
    while(b):
        q, a, b = a//b, b, a%b
        r0, r1 = r1, r0 - q*r1
        s0, s1 = s1, s0 - q*s1
    return a, r0, s0
```

Analysis of The Extended Euclidean Algorithm.

- Correctness: it is easy.
- Efficiency: it is trivial.
- Optimization: binary-egcd.

Concluding Remarks.

- The content is easy, but is not trivial.
- Algorithms should be focused by CSers.
- Useful tools should be mastered.

Happy ending.

- Any questions?
- Feedback is welcome: lbwang@gmail.com.
- Thanks!